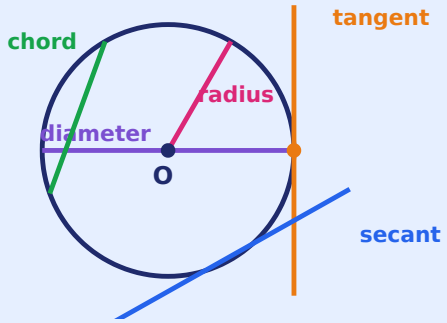
Unit 9 · Circles Cheat Sheet

Unit 9 focus: parts of a circle, circumference, area, arcs and sectors

1 KEY VOCABULARY

radius	Center to a point on the circle.	r
diameter	Chord through the center: $d = 2r$.	$d = 12$
chord	Segment with both endpoints on the circle.	$\overline{AB}$
tangent	Touches the circle at exactly one point.	$\perp$ radius
secant	A line through two points of the circle.	
arc	Part of the circle between two points.	$\overset{\frown}{AB}$

2 PARTS OF A CIRCLE



3 CIRCUMFERENCE

$C = 2\pi r = \pi d$: the distance around.

Example: Radius 5
Formula: $C = 2\pi(5) = 10\pi$
Answer: $10\pi \approx 31.4$ **units**

4 AREA

$A = \pi r^2$: the space inside.

Example: Radius 6
Formula: $A = \pi(6)^2 = 36\pi$
Answer: $36\pi \approx 113.1$ **units²**

5 ARC LENGTH & SECTOR AREA

Find	Formula	Example: $60°$, $r = 9$
arc length	$\frac{\theta}{360°} \cdot 2\pi r$	$\frac{1}{6}(18\pi) = 3\pi$
sector area	$\frac{\theta}{360°} \cdot \pi r^2$	$\frac{1}{6}(81\pi) = 13.5\pi$

6 QUICK REMINDERS & COMMON MISTAKES

✓ Do remember

- Radius is half the diameter.
- Use degrees in the arc and sector fractions.
- Leave answers in terms of π unless told to round.

✗ Common mistakes

- Using the diameter in πr^2: halve it first
- Circumference vs area: distance around vs space inside
- Arc measure vs arc length: degrees vs a length

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More practice: inscribed angles, chords, tangents and angles inside and outside a circle

1 CENTRAL & INSCRIBED ANGLES

Angle	Measure
central (vertex at center)	= its arc
inscribed (vertex on circle)	$\frac{1}{2}$ its arc
inscribed in a semicircle	90°
tangent and chord	$\frac{1}{2}$ its arc

Arc 80°: central 80°, inscribed 40°.

2 CHORDS

- Congruent chords $\leftrightarrow$ congruent arcs.
- A radius perpendicular to a chord bisects the chord and its arc.
- Chords that cross inside: $a \cdot b = c \cdot d$ (products of the pieces).

3 TANGENTS

- A tangent is perpendicular to the radius at the point of tangency.
- Two tangent segments from the same outside point are congruent: $PA = PB$.
- Use the Pythagorean Theorem with the radius, tangent and the segment to the center.

4 ANGLES INSIDE THE CIRCLE

Two chords crossing: $m\angle = \frac{1}{2}(\text{arc}_1 + \text{arc}_2)$.

Example: Intercepted arcs 80° and 100°
Average: $\frac{1}{2}(180^\circ)$

Answer: 90°

5 ANGLES OUTSIDE THE CIRCLE

Secants or tangents meeting outside: $m\angle = \frac{1}{2}(\text{far arc} - \text{near arc})$.

Example: Arcs 140° and 60°
Half the difference: $\frac{1}{2}(80^\circ)$

Answer: 40°

6 EQUATION OF A CIRCLE

Center (h, k), radius r: $(x-h)^2 + (y-k)^2 = r^2$.

Example: Center $(2, -3)$, radius 4
Substitute: $(x-2)^2 + (y+3)^2 = 16$

Answer: $(x-2)^2 + (y+3)^2 = 16$